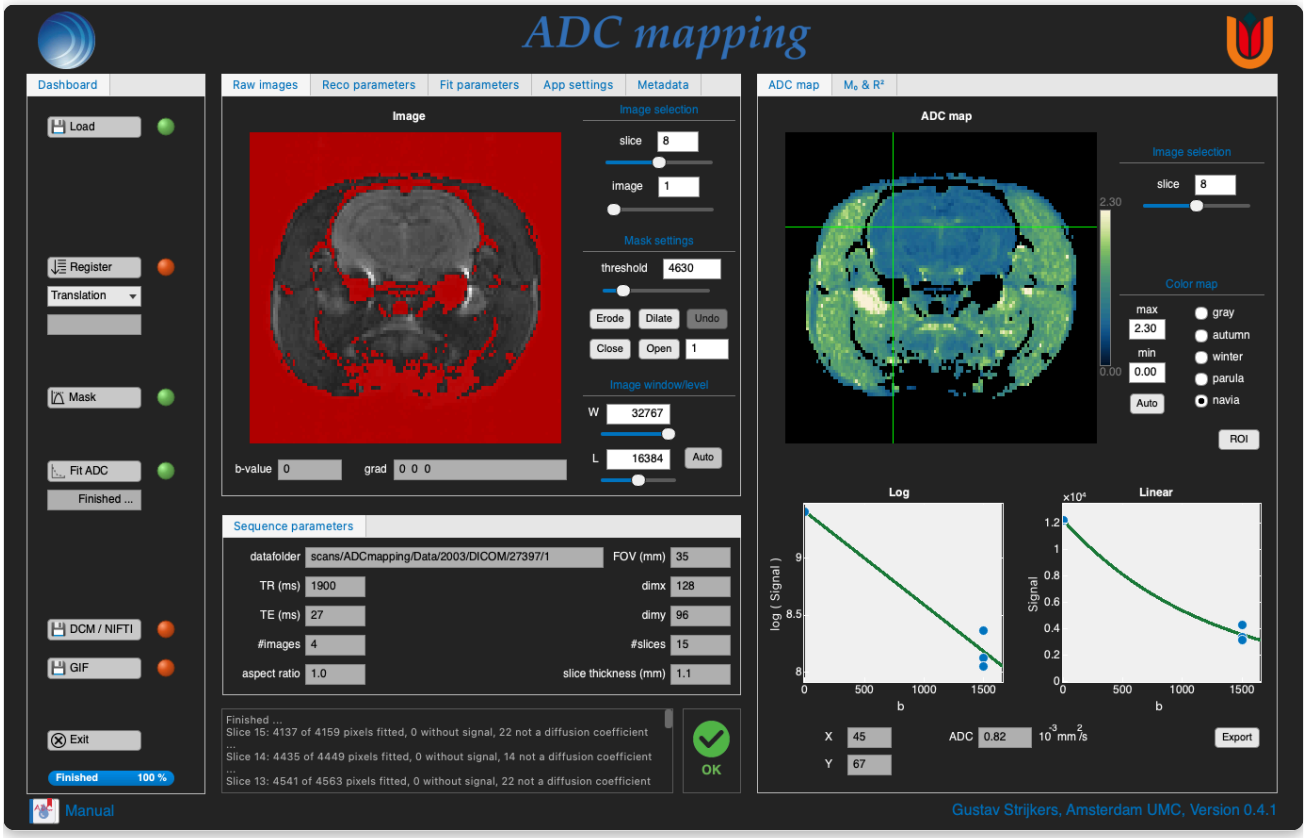


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1. Overview

ADCmapp is an application with a graphical user interface to reconstruct apparent diffusion coefficient (ADC) maps from diffusion-weighted acquisitions with an MR Solutions preclinical MRI system. It reads a diffusion-weighted EPI scan from its MRD files, which it reconstructs, or a series of DICOM files. The images are registered, masked and fitted to ADC, M_0 and R^2 maps, which can be exported as DICOM, NIfTI and GIF, with the values of a region of interest as a CSV file.

Note: the program does not calculate a diffusion tensor, but only estimates the mean diffusivity / ADC.

ADCmapp is a stand-alone application: everything it uses, elastix and dcm2niix included, is inside it, and nothing else has to be installed.

2. Contact

ADCmapp and this manual are written by:

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3. Disclaimer

This is a beta version. The software has been extensively tested using mouse and rat data acquired with an MR Solutions preclinical 7.0T/24cm system. However, this does not warrant that the functions contained in the program will meet your requirements or that the operation of the program will be uninterrupted or error-free. In case of questions or issues, please contact Gustav Strijkers or Bram Coolen.

4. Installation

macOS

Requirements: a Mac with Apple silicon (M1 or later), macOS 26 or later.

ADCmapp comes as a disk image, `ADCmapp-0.4.7-macOS-arm64.dmg`. Nothing else has to be installed: Python, elastix, dcm2niix and everything else the app uses are inside it.

1. Open the disk image and drag **ADCmapp** onto the Applications folder in its window.
2. Open it from Applications. The app is not signed with an Apple Developer ID, so the first time macOS refuses it ("Apple could not verify ..."); click **Done**. Then open **System Settings > Privacy & Security**, scroll down to the line saying ADCmapp was blocked, click **Open Anyway** and confirm. From then on it opens as any other app.

   Instead of step 2, the same in Terminal, once:

   ```
   xattr -dr com.apple.quarantine /Applications/ADCmapp.app
   ```

The first start can take a little longer: macOS checks the app once.

Windows

Requirements: a PC with Windows 10 or 11, 64-bit.

ADCmapp comes as an installer, `ADCmapp-0.4.7-setup.exe`. Nothing else has to be installed: Python, elastix, dcm2niix and everything else the app uses are inside it.

1. Run the installer. It is not signed, so the first time Windows may stop it ("Windows protected your PC"): click **More info** and then **Run anyway**.
2. The installer puts the app in `%LOCALAPPDATA%\Programs\ADCmapp` for the current user, without administrator rights (installing it for all users is an option), with a Start-menu entry and, if you choose, a desktop shortcut. At the end it can start the app.

To remove the app, use **Uninstall ADCmapp** in the Start menu, or **Settings > Apps**.

5. Running the software

Open **ADCmapp** from the Applications folder (macOS), or from the Start menu or the desktop shortcut (Windows). The ADCmapp logo appears while the app loads, and the window follows within seconds. At its start the app looks for elastix, and the message box reports what it found; "Ready" says it is done. Only **Load** and **Exit** are on until a dataset is loaded.

A scan given on the command line (an MRD or DICOM file or its folder, as a program that starts ADCmapp with a scan gives it) is loaded at once, as with **Load**; a path the app cannot load gives a warning in the message box. When the app is open already, a scan given this way goes to the open window instead: the window comes to the front and loads it, or loads it when the step under way ends.

Newer versions. At its start the app asks GitHub (https://github.com/Moby1971/MRI-apps) whether there is a newer release of it. The message box then shows the current and the newest version and, when there is a newer one, where to download it. Without a network connection it says that the version could not be checked, and the app works as usual. `checkForUpdates` in the settings file switches the check off; `versionCheckTimeout` sets how many seconds it waits for GitHub (5).

Updates (App settings tab):

- **auto update**, ticked as it comes: new versions of ADCmapp arrive by themselves. When one is ready, a window says so with what is new: **Install and Restart** installs it and starts the new ADCmapp; **Later** installs it when you quit ADCmapp, without asking again. The window waits while ADCmapp is working. Without an internet connection nothing is said and nothing waits. On Windows, ADCmapp installed for all users can be updated only by an administrator; for other users the message box says once where to get the new version to install by hand.
- **Check now** looks for a newer version at once, also with **auto update** off; the message box says what it found. On the Mac, **Check for Updates...** in the ADCmapp menu does the same.
- **notify at start**, ticked as it comes, counts only with **auto update** off: when a newer version is available, a window at the start offers it. **Download** gets the installer for this computer, **Skip this version** shows no window for that version again, **Later** asks again at the next start.

Moving to MAPapp. ADCmapp continues as MAPapp, one app for the quantitative maps, which has the diffusion (ADC) method. MAPapp starts with its own settings. Once MAPapp is released, ADCmapp says so in a window at its start: "ADCmapp continues as MAPapp." **Install MAPapp now** downloads and installs MAPapp, moves ADCmapp to the Trash (on Windows: uninstalls ADCmapp) and opens MAPapp; **Later** keeps ADCmapp as it is and asks again at the next start. When MAPapp is already installed on your computer, the window says so instead and offers **Open MAPapp**, which opens it and quits ADCmapp (ADCmapp stays installed), or **Later**. The window comes with the check for new versions: with **auto update** on, or with **Check now**; it does not come while a step runs.

The message box says at every start whether ADCmapp is up to date.

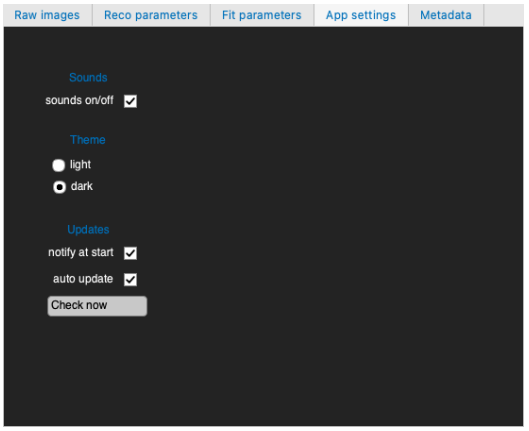
This manual. The **Manual** button at the bottom of the window, with the app's icon on a book, opens this manual in the computer's PDF viewer.

The window has a fixed size. On a screen too small for it, the window fits the screen and can be scrolled.

Settings and log files

ADCmapp keeps its files in a folder of its own, which it makes at its first start:

macOS	`~/Library/Application Support/ADCmapp/0.4.7p`
Windows	`%APPDATA%\ADCmapp\0.4.7p`



Settings. The settings are kept in `adcmappSettings.json`. The file is written with the standard values the first time a version starts, and can then be edited by hand; changes take effect at the next start. It holds the version check, the folders where loading, the gradient table and the exports start (`importPath`, `gradientTablePath`, `defaultExportPath`), the theme, the sounds, the corrections of the MRD reconstruction, MP-PCA de-noising (`denoising.pca`, on, and `denoising.dicom`, off: also for DICOM series), the registration method, the R^2 cut-off and whether it is applied, the color maps of the three maps, and the number of log files kept. A value that cannot be used is replaced by its standard value, and the message box says so; a file that cannot be read at all is set aside as `adcmappSettings-unreadable-<time>.json` and replaced by a new one. The App settings tab switches the theme, the sounds, notify at start and auto update.

ADCmapp keeps your choices of the theme, the sounds, notify at start, auto update, the color maps, the registration method, the R^2 cut-off and whether it is applied, and the reconstruction's corrections, and the folders last used for loading and for exports, for the next start; a new version takes them over.

Log files. Every session writes a log file, `adcmapp-log-<date and time>.txt`, to the same folder. The log keeps every message and the parameters of each load, registration and fit. Logs older than the newest 50 are removed at start.

6. Basic operation



The ADCmapp program operates from a single window with 5 panels.

Panel 1: Dashboard. The task buttons and parameters that control the process. The tasks run from top to bottom; registration and masking can be left out. A green light shows a task done, yellow a task running, and red one not done or failed. At the bottom of the Dashboard one progress bar shows the progress of every task ("Loading 40 %", "Finished").

Panel 2: Raw images / Reco parameters / Fit parameters / App settings / Metadata. The last tab, **Metadata**, holds the record of the scan: the animal, the session and the contrast agent (step 8).

Panel 3: Sequence parameters. The relevant acquisition parameters, and the dataset loaded (**datafolder**).

Panel 4: ADC map / M_0 & R^2. During and after fitting, the ADC maps. The M_0 and R^2 goodness-of-fit maps are displayed in the second tab.

Panel 5: Messages. The program messages, newest on top, and the status: OK, BUSY, WARNING or ERROR.

Step 1: Loading data

Press **Load** and choose a file:

- an **MRD file** of a diffusion-weighted EPI scan, which is then reconstructed (step 2), or
- **any DICOM file of a series**: the series in its folder is loaded.

The b-value and gradient direction of every image of a DICOM series come from its files. When the scan's MRD file lies where the scanner keeps it (the MRD folder of the same exam and scan number), they come from the scan's protocol instead, and a warning says so when the files give other values. Without the MRD file, a warning names any image whose b-value its signal contradicts: an image with a large b-value that is as bright as the images without diffusion weighting. Check the b-values of such a series, and load a gradient table that gives the right ones (**Import**, step 5).

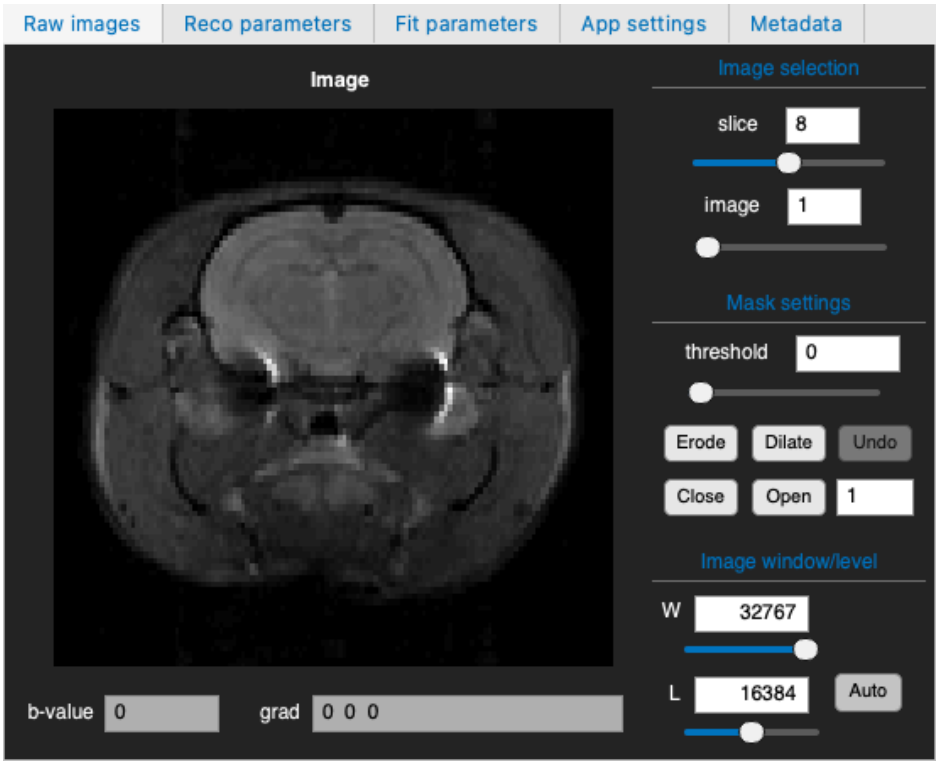
On the scanner computer (Windows) the first dialog of a session opens in the newest exam of the scanner's Data folder, where its DICOM series are, unless the settings name a folder (`importPath`). Elsewhere it opens in the folder of the last file loaded.

The middle slice and the first image are shown. A dataset that cannot be used is refused, with the reason in the message box, and the dataset loaded before stays. While a load runs, the **Load** button reads **Stop**; pressing it stops the load, and the dataset loaded before stays as it was. Reconstructions written by Retrospective or P2ROUD are not measurements and are refused: choose the scan's own MRD file.

The relevant acquisition parameters are shown in panel 3.



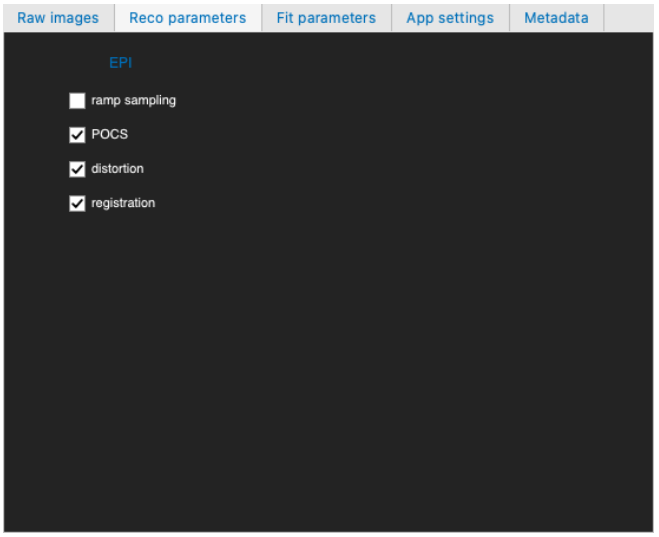
Images with different b-values and diffusion directions can be inspected with the sliders of the Raw images tab (**slice** and **image**); the **b-value** and gradient direction (**grad**) of the image are shown below it. The window and level are set with the **W** and **L** fields and sliders, or with **Auto**. The mouse wheel over a slider moves that slider, one step per notch.



Step 2: MRD scans

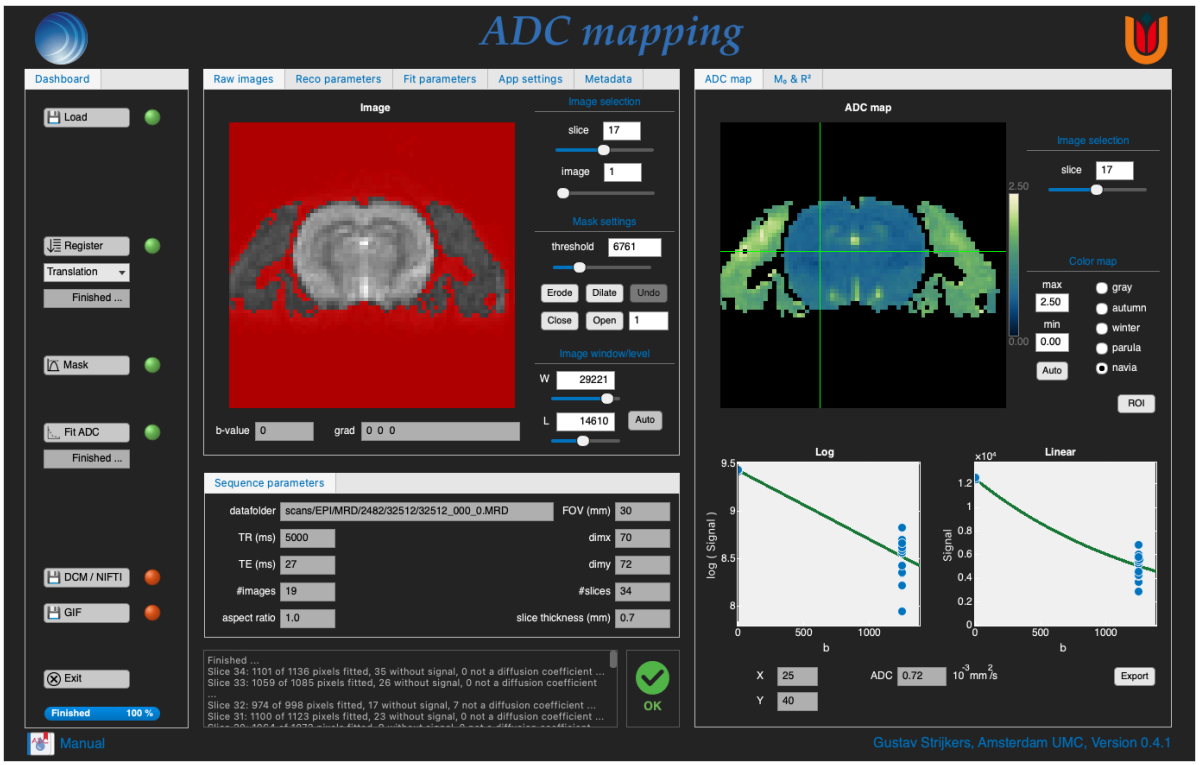
An MRD scan must be a 2D diffusion EPI with at least two b-values. Its MRD files are reconstructed at load, with the corrections of the Reco parameters tab:

- **ramp sampling** (off by default) — also uses the samples taken on the readout gradient ramps;
- **POCS** — completes a partial Fourier EPI, which sharpens the image and costs signal-to-noise ratio;
- **distortion** — the EPI geometric distortion is corrected, where the study holds a scan of the same volume with the phase encoding reversed;
- **registration** — the volumes of the series are aligned.



They start as the settings file says and take effect at the next load; MP-PCA de-noising is set in the settings file. The images are masked when the study has a FLASH mask scan. The reconstruction shows its progress on the progress bar; while it runs, **Load** reads **Stop**, and pressing it stops the reconstruction and leaves the dataset loaded before and the settings as they were.

Loading writes nothing beside your data. The **DCM / NIFTI** export writes the reconstruction (step 9). The window below shows a DTI scan of a brain, loaded from its MRD files, registered and fitted.



Step 3: Image registration

NOTE: this step usually can be omitted.

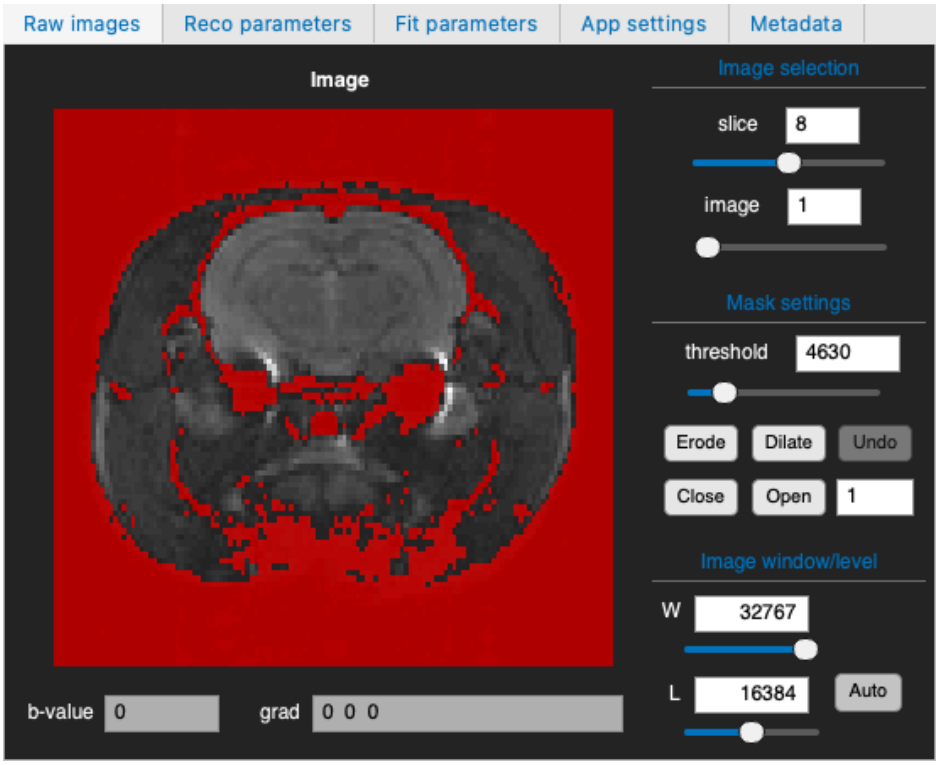
In case the images with different b-values and directions are not perfectly registered, press **Register** to register every image to the first image with the lowest b-value of its slice, with elastix, as a **Translation**, **Rigid**, **Affine** or **B-Spline** transform (the list below the button). While it runs, the button reads **Stop**; pressing it stops the registration and leaves the images as they were. Registering clears the mask and the maps. A series with many images takes a while: the DTI scan above, 19 images in 34 slices, took several minutes.

Step 4: Masking the background

To prevent unnecessary fitting of the background pixels, the images need to be masked first; without a mask every pixel is fitted, the background included, with a warning.

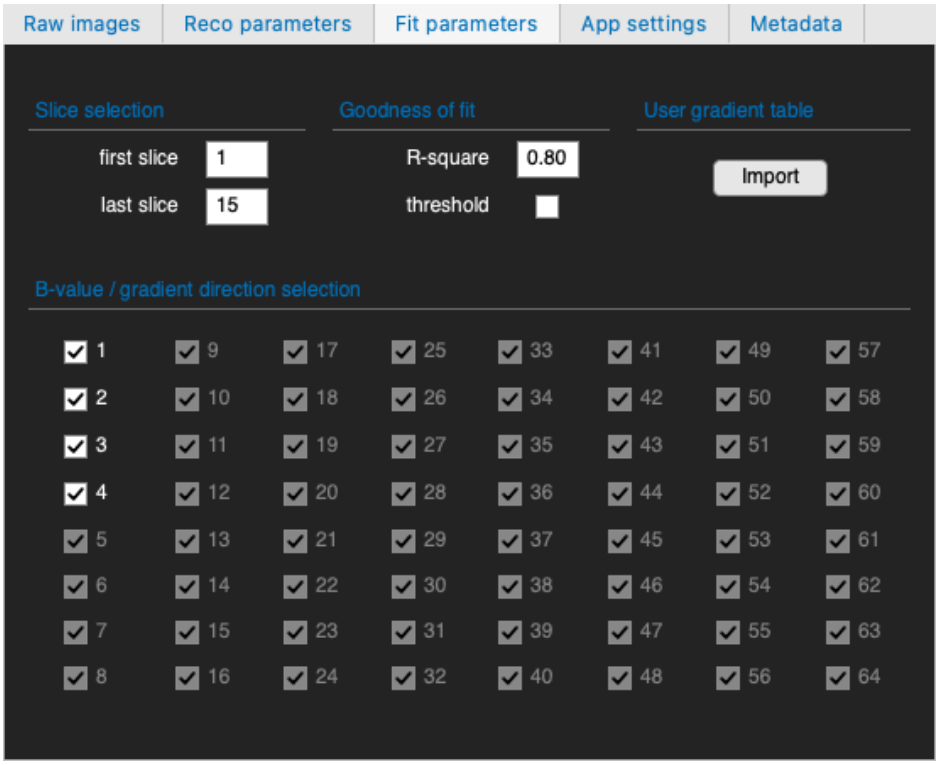
Press **Mask** to mask them with an automatic (Otsu) threshold per slice, on the first image with the lowest b-value. The red pixels indicate the regions that will be skipped during ADC fitting.

In case the automatic threshold is not satisfying, it can be adjusted per slice with the **threshold** field or slider. The mask can be refined by **Erode**, **Dilate**, **Close** and **Open** operations, of the size in pixels in the field beside them; **Undo** takes the last one back. Any change to the mask after a fit, **Mask** included, withdraws the fitted maps: fit again to see and export them.



Step 5: Fitting ADC maps

The Fit parameters tab holds the fit settings.



The first and last slice that will be fitted can be selected.

When the **threshold** box is ticked, the ADC and M_0 of a pixel whose R^2 is below **R-square** (0.80 by default) are cleared; the R^2 map keeps them, so the rejected pixels can be seen.

R^2 measures how far the decay over the b-values departs from a mono-exponential.

R^2 needs at least 3 different b-values. With two — b = 0 and one shell, the usual trace or DTI acquisition — R^2 is 1 in every pixel and the threshold removes nothing; the message box says so when the fit starts.

The bottom part of this tab lists all the images (b-value and diffusion direction combinations). The images that are ticked are included in the fit.

Import reads a gradient table: a text file with a line per image, the b-value (s/mm²) and the three components of the direction, separated by spaces, tabs, commas or semicolons; % and # start comments. A dataset has to be loaded first; maps fitted before need to be fitted again to use the table.

Press **Fit ADC** to start the ADC fitting; while it runs, the button reads **Stop**, and pressing it stops the fit and leaves the maps fitted before. The maps fill in slice by slice. The selected b-values must take two values or more, at least 100 s/mm² apart. The ADC, in 10^{-3} mm²/s, is fitted to:

> $Signal = M_0 \exp(-b \cdot ADC)$

For each slice the message box says how many pixels were fitted.

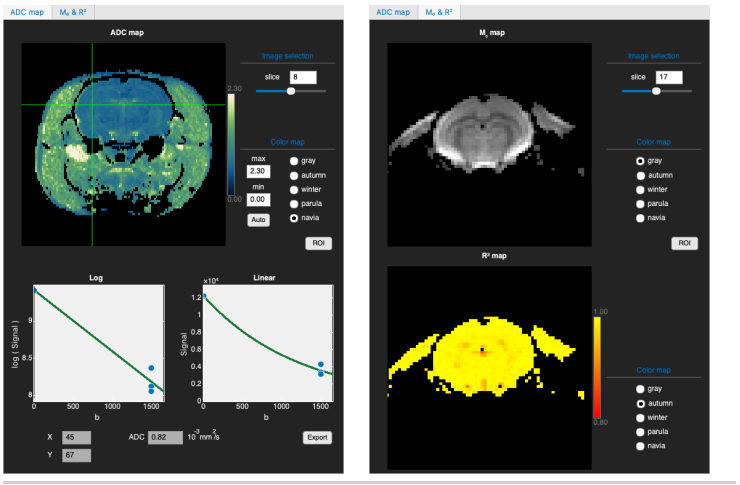
The acquisition. The fitted ADC is the mean diffusivity (the trace of the diffusion tensor over three) when each b-value is acquired with a balanced set of directions: three orthogonal directions, or a set spread evenly over the sphere, as a DTI scheme is. With other schemes the ADC leans toward the directions measured, and so does R^2; a scan meant for ADC maps should be planned that way.

Step 6: Fit results

The ADC maps are shown in the ADC map tab, on the scale set with **max** and **min**, or with **Auto** (0 to twice the mean ADC). An M_0 map and an R^2 map are shown in the second tab, each with its own color map (gray, autumn, winter, parula, navia).

The color bars stand beside the ADC and R^2 maps, with their end values above and below them: the scale's max and min for the ADC, 1 and the R^2 cut-off for R^2.

ADC map **M_0 & R^2, multi-shell DTI**

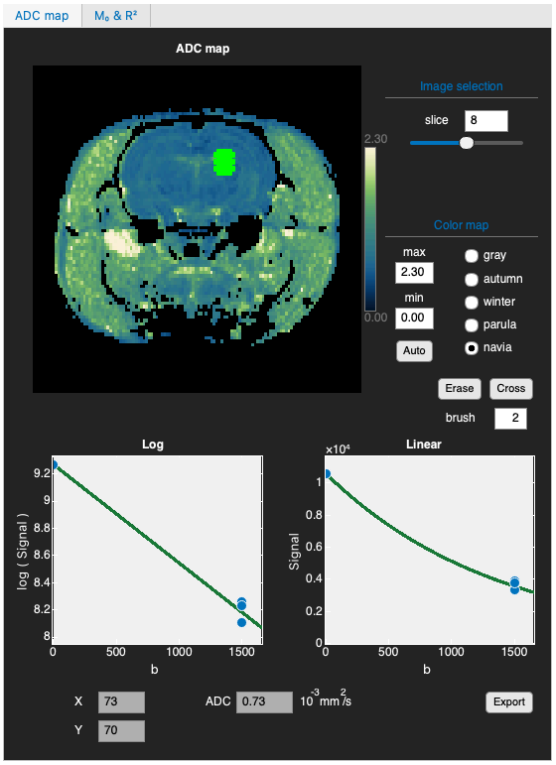


The M_0 & R^2 example is a multi-shell DTI scan (b = 0, 1000 and 2000 s/mm²): with three b-values R^2 shows where the decay is not mono-exponential.

Step 7: ADC values of pixels and ROIs

When you click on the ADC map (or the M_0 map) a cross-hair is shown, with the log(signal) and the signal of that pixel as a function of b (symbols), the fitted curve (line) and its ADC. A red symbol indicates that this image was left out of the fit, according to the list in the Fit parameters tab (see step 5).

You can toggle between the cross-hair and region-of-interest (ROI) values with the **ROI** / **Cross** button. A single ROI can be 'painted' on the map, in one or more slices, by mouse clicks on the image, with the brush size in pixels set below the button. The plots then show the mean signal of the ROI and the curve of its mean ADC and M_0.

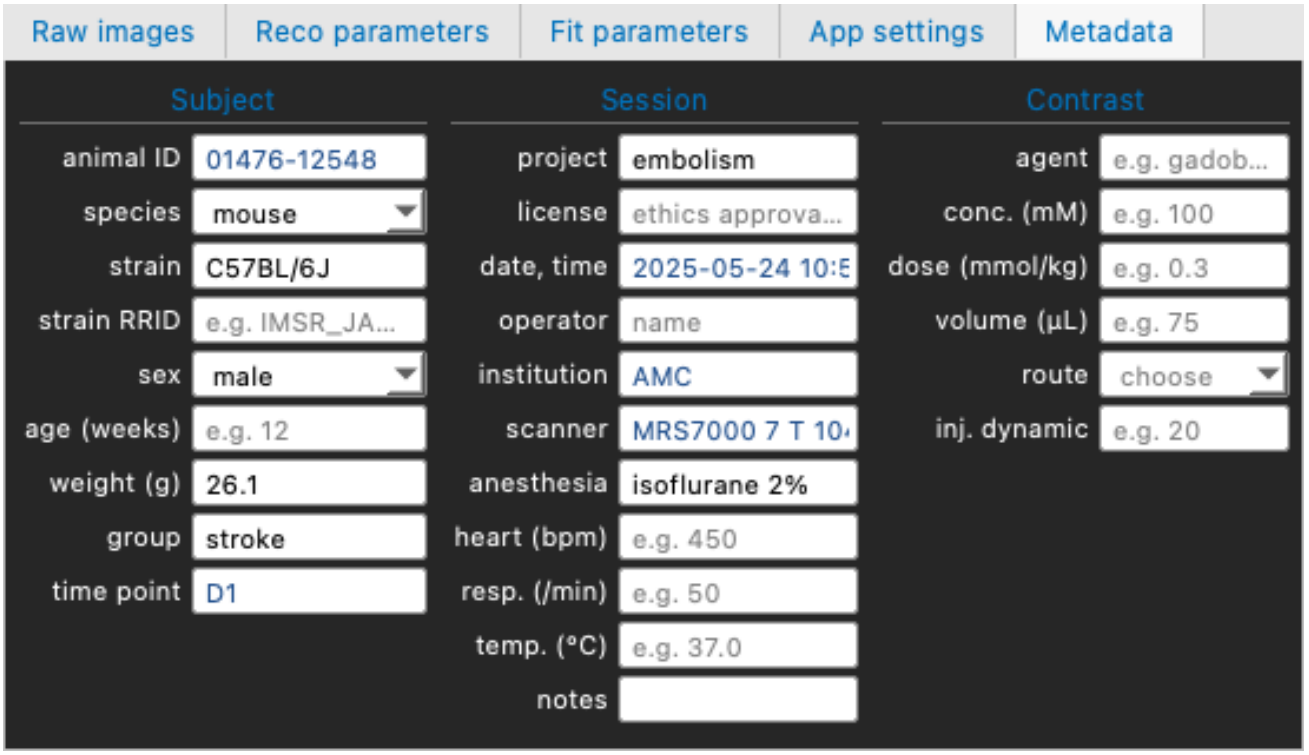


The ROI can be removed with **Erase**. **Export** writes the b-values and the signal under the cross-hair or of the ROI to a CSV file, `roiData_<tag>.csv`, in a folder you choose.

Step 8: Metadata

The **Metadata** tab, beside the Raw images, Reco parameters, Fit parameters and App settings tabs, holds the record of the scan, which every export writes with the maps. Its fields stand in three columns:

- **Subject:** animal ID, species, strain and its RRID, sex, age, weight, group and time point;
- **Session:** project, ethics license, date and time, operator, institution, scanner, anesthesia, heart and respiration rate, and temperature, with a field for notes;
- **Contrast:** agent, concentration, dose, volume, route and the dynamic it was injected at.



At a load the app fills in what the scan's files say: the animal ID and the time point from the patient name typed at the console (`20240212_926752_D0`: animal 926752, day 0), the project, the date and time, the institution and the scanner. What the console leaves at its standard values (sex O, weight 0, the login as the operator) is left empty. A value the app filled in is shown in blue; every field can be corrected, and then counts as yours.

Your institution, operator, anesthesia, species, strain and strain RRID are kept for later exams: they are shown, in blue, from the start, and a scan's own value (from its files or its exam) takes their place where it has one.

What was filled in for an earlier scan of the same exam is filled in again (ADCmapp and the other apps alike): the animal's details are typed only once per exam. The date, the heart and respiration rate, the temperature, the notes and the injection's dynamic belong to one scan and are never taken from another.

Species, sex and route are lists to choose from, or to type in; numbers are kept with their units (a number field holding something else is edged in red). The species and sex are written with their codes (NCBI Taxonomy, PATO), so the record can be read by other software as well as by people.

Step 9: Exporting the maps

There are two ways to export the ADC, M_0 and R^2 maps.

1. **DCM / NIFTI** — exports the ADC, M_0 and R^2 maps in DICOM and NIfTI format for further processing in third-party software, with a copy of the diffusion-weighted images. For an MRD scan it first writes the reconstruction: the DICOM series with `recoparameters_<tag>.txt`, its NIfTI files with the b-values and directions (`.bval`, `.bvec`), scripts for MRtrix3 and DSI Studio where the diffusion scheme allows it (written, not run), and a BIDS copy. The reconstruction and its maps take their field of view, pixel size and position from the scan's own parameters, so that

they lie where the scan was planned. The DICOM maps, and the reconstruction of an MRD scan, carry the species, strain, sex, age and weight of the animal and the contrast agent, where the Metadata tab has them.

2. **GIF** — exports the three maps as animated GIF files, one frame per slice, with two pictures of the app's window.

The exports go beside the scanner's files, in …/Data/<study>/DICOM, Nifti, BIDS and GIF, when the dataset was loaded from the scanner's folders; otherwise the app asks for a folder, and starts at the folder chosen last. Every export gets a new numbered folder, so that no export overwrites another.

The export folders and files are named by the series number or the scan's name (<tag>); spaces and other special characters in it become _, so that the names have no spaces.

Every export also writes, in …/Data/<study>/Metadata/<tag>ADC/<n>:

- metadata.json — the Metadata tab's record;
- adcmapp-report_<tag>.json and .txt — the report of the export: the app and its libraries, the scan and a checksum of its files, its acquisition with the b-values and directions, every step of the session with its parameters (the reconstruction of an MRD scan, the gradient table, the registration, the mask and your changes to it, the fit with all its settings), the statistics of each map per slice, the b-values the fit used, and every file the export wrote with its checksum;
- ro-crate-metadata.json — the same as an RO-Crate (https://www.researchobject.org/ro-crate/), a standard description of the folder for data repositories.

The paths in them are relative to the study folder, so they hold no names of your own folders.

Step 10: Exit

Press **Exit** to shut down the program. Closing the window does the same, also while a step runs: the step is stopped, the window disappears, and the program closes as soon as the step has ended.

7. Credits

- **elastix** — S. Klein, M. Staring, K. Murphy, M.A. Viergever, J.P.W. Pluim, elastix: a toolbox for intensity based medical image registration, IEEE Transactions on Medical Imaging 29(1), 196–205 (2010). https://elastix.dev
- **PCA denoising** — J. Veraart, D. S. Novikov, D. Christiaens, B. Ades-Aron, J. Sijbers, E. Fieremans, Denoising of diffusion MRI using random matrix theory, NeuroImage 142, 394–406 (2016).
- **Color maps** — F. Crameri, Scientific colour maps (navia), Zenodo, doi:10.5281/zenodo.1243862.
- **NIfTI conversion** — dcm2niix: C. Rorden et al., https://github.com/rordenlab/dcm2niix.